

# CE 310 — Week 5 Assignment: Normal Distribution Applications

Due: Wednesday of Week 6 by 9:00 AM — submit to D2L Total Points: 60 pts

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## Overview

This assignment extends the Week 5 lab into new engineering scenarios. You will apply the Normal distribution parameters you computed in the lab — concrete compressive strength and vehicle speed — to thresholds and percentiles not covered in the lab, interpret the results through an engineering design lens, and explain in writing what the distribution reveals that the mean alone cannot.

All three problems use parameters you already computed: Week5\_CE\_Measurements.csv

Do not re-derive  $\mu$  and  $\sigma$  from scratch. Reference your lab Answer Sheet cells directly in your formulas.

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## How to Submit

☐ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

| Requirement   | Detail                                                                                           |
|---------------|--------------------------------------------------------------------------------------------------|
| File format   | .xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF               |
| Sheet name    | Must be exactly Answers — do not rename or delete it                                             |
| Column layout | Enter numeric answers and written responses in Column C — do not insert, delete, or move columns |
| File name     | CE310_W5_<NetID>.xlsx<br>(e.g. CE310_W5_abc123.xlsx)                                             |
| Submission    | Upload the .xlsx file to D2L Dropbox before the deadline                                         |

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## Problem 1 — New Concrete Strength Probabilities (30 pts)

Use the all-specimen Normal model from your lab:  $\mu = A1$  ( $\approx 4,496$  psi),  $\sigma = A2$  ( $\approx 442$  psi).

These problems use thresholds different from those in the lab. Do not reuse lab thresholds.

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### Part 1a — High-Strength Classification (10 pts)

In ASTM and industry practice, concrete achieving 5,200 psi or more is often classified as “high-strength” concrete. Compute:

$$P(f'c > 5,200 \text{ psi})$$

The fraction of specimens from this mix that would qualify as high-strength.

Show your formula and result. Then write 2–3 sentences interpreting this value: What does this percentage tell a materials engineer about how reliably their mix achieves high-strength performance? Include the z-score for 5,200 psi in your interpretation.

Enter your written response in the P1a\_written cell of your Answer Sheet.

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### Part 1b — Target Zone Probability (10 pts)

A QC engineer wants to know how often batches land in the “target zone” — neither too weak nor over-designed. Compute:

$$P(4,500 \text{ psi} < f'c < 5,000 \text{ psi})$$

Show your formula and result. Then write 2–3 sentences interpreting this value: What fraction of batches land in this target zone? Is this range wider or narrower than you would expect based on the 68–95–99.7 rule, and why?

Enter your written response in the P1b\_written cell of your Answer Sheet.

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### Part 1c — Eurocode 7 Characteristic Strength (10 pts)

Eurocode 7 defines the characteristic value of a material property as its 5th percentile — the value that 95% of specimens are expected to exceed. Compute:

The 5th percentile concrete compressive strength

Show your formula and result. Then write 2–3 sentences interpreting this value: How does the 5th percentile compare to the specified design strength of  $f'c = 4,000$  psi? What does

this imply about the suitability of the all-specimen mix design for a project governed by the Eurocode 7 characteristic value criterion?

Enter your written response in the P1c\_written cell of your Answer Sheet.

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## Problem 2 — Traffic Speed Compliance Analysis (18 pts)

This problem uses the three speed Normal models from your lab. Reference your lab Answer Sheet cells directly.

- All speeds:  $\mu = B1$  ( $\approx 54.4$  mph),  $\sigma = B2$  ( $\approx 7.8$  mph)
  - AM Peak:  $\mu = B8$  ( $\approx 52.2$  mph),  $\sigma = B9$  ( $\approx 8.2$  mph)
  - PM Peak:  $\mu = B10$  ( $\approx 57.2$  mph),  $\sigma = B11$  ( $\approx 6.2$  mph)
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### Part 2a — Excessive Speed Exceedance (6 pts)

In some federal highway jurisdictions, 65 mph is the threshold used to flag “excessive speed” for enforcement analysis. Compute  $P(\text{Speed} > 65 \text{ mph})$  separately for all three distributions:

- All speeds (use  $\mu = B1$ ,  $\sigma = B2$ )
- AM Peak only (use  $\mu = B8$ ,  $\sigma = B9$ )
- PM Peak only (use  $\mu = B10$ ,  $\sigma = B11$ )

Show all three formulas and results. In 1–2 sentences, identify which period has the highest exceedance fraction and suggest one reason why AM Peak would produce a higher rate than PM Peak given the same speed limit.

Enter your written response in the P2a\_written cell of your Answer Sheet.

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### Part 2b — 90th Percentile Design Speed (6 pts)

Traffic engineers sometimes use the 90th percentile speed as a more conservative design speed criterion than the traditional 85th percentile. Compute the 90th percentile for all three distributions:

- All speeds (use  $\mu = B1$ ,  $\sigma = B2$ )
- AM Peak only (use  $\mu = B8$ ,  $\sigma = B9$ )
- PM Peak only (use  $\mu = B10$ ,  $\sigma = B11$ )

Show all three formulas and results. Then write 2–3 sentences: How do these 90th percentile values compare to the 85th percentile speed you computed in the lab (B7)? What

is the practical difference between using the 85th versus the 90th percentile as the design speed criterion, and which is more conservative?

Enter your written response in the P2b\_written cell of your Answer Sheet.

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### Part 2c — Evaluating a Speed Limit Argument (6 pts, written response)

A traffic engineer argues:

“The mean speed on this corridor is 54.4 mph — only 9.4 mph over the posted 45 mph limit. A 10 mph speed limit increase to 55 mph would resolve the compliance issue.”

Write 3–4 sentences that evaluate this argument using your results from P2a and P2b. Does raising the posted limit to 55 mph address the exceedance problem revealed by the 85th percentile? Use at least two specific numbers from P2a or P2b in your response.

Enter your written response in the P2c\_written cell of your Answer Sheet.

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### Problem 3 — Reflection (12 pts)

A project manager reviews the concrete QC report and says:

“Both the Alpha and Beta suppliers show average strength well above 4,000 psi — we don’t need to worry about which supplier we use.”

Write one paragraph (100–150 words) that explains why this reasoning is incomplete. Your response must:

1. Reference the ACI 318 10th percentile rule you applied in the lab
2. Cite the specific 10th percentile values for Alpha ( $A_7 \approx 4,191$  psi) and Beta ( $A_9 \approx 3,747$  psi)
3. Explain what  $\sigma$  tells you about mix performance that the mean does not

Enter your reflection in the P3\_written cell of your Answer Sheet.

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### Submission Requirements

- Type all written responses in the designated yellow cells of your Assignment Answer Sheet. Submit CE310\_W5\_<NetID>.xlsx to D2L.
  - Show all Excel formulas with the formula bar visible for each NORM.DIST or NORM.INV expression.
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## Grading Summary

| Problem                                                                                                                   | Points |
|---------------------------------------------------------------------------------------------------------------------------|--------|
| P1a — $P(f'c > 5,200 \text{ psi})$ : formula, result, 2–3 sentence interpretation with z-score                            | 10     |
| P1b — $P(4,500 < f'c < 5,000)$ : formula, result, 2–3 sentence comparison to 68–95–99.7 rule                              | 10     |
| P1c — 5th percentile strength: formula, result, 2–3 sentence Eurocode 7 interpretation                                    | 10     |
| P2a — $P(\text{Speed} > 65 \text{ mph})$ for all / AM / PM: all three formulas and results, 1–2 sentence written response | 6      |
| P2b — 90th percentile speed for all / AM / PM: all three formulas and results, 2–3 sentence comparison to 85th pct        | 6      |
| P2c — Speed limit argument evaluation (3–4 sentences, $\geq 2$ specific numbers cited)                                    | 6      |
| P3 — Reflection paragraph (100–150 words; ACI 318 rule cited; A7 and A9 values cited; $\sigma$ explained)                 | 12     |
| Total                                                                                                                     | 60 pts |

This assignment is worth 60 raw points and will be scaled to the Post-Lab Memo category in the course gradebook.